

Fast metamaterial design optimization using reduced order unit cell modeling

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Abstract

Metamaterials have recently emerged and shown potential as lightweight and compact noise and vibration solutions. By adding or embedding resonant structures to or in a flexible host structure on a sub-wavelength scale, stop bands can be created, which enable frequency ranges of strong vibration and noise attenuation. These often periodic structures are typically analyzed using infinite periodic structure theory, based on a representative unit cell model. The stop bands in metamaterials are generally analyzed by means of dispersion curves. For complex unit cell models, however, this rapidly becomes computationally expensive. Therefore, in this work, model order reduction is applied to finite element unit cell models for stop band predictions. In view of designing metamaterials for a targeted frequency range, the reduced order modeling approach is next introduced in a design optimization routine. The results show that the reduced order models for stop band predictions enable substantial gains in metamaterial analysis and fast metamaterial design optimization.

1 Introduction

In recent years, metamaterials have emerged as potential lightweight and compact solutions with favorable noise and vibration attenuation performance in targeted frequency ranges. Through the addition of structural resonators to or in a flexible host structure on a sub-wavelength scale, a resonance-based stop band for structural wave propagation can be obtained around the tuned resonator frequency [1, 2]. A resonance-based stop band for the acoustically relevant bending waves in partitions can also enable a targeted frequency range of strongly increased sound transmission loss, outperforming the acoustic mass-law [1, 3]. The improved noise and vibration attenuation potential of these vibro-acoustic metamaterials has been experimentally demonstrated for a variety of configurations, ranging from metamaterial beams [4–6], ducts [7] and pipes [8] to metamaterial plates [9–12], double panels [13, 14], sandwich panels [15] and honeycomb panels [16].

When predicting the vibro-acoustic performance of metamaterials, the primary interest usually lies in analyzing their stop band behavior. To this end, the often, yet not strictly necessary, periodic nature of metamaterials is generally exploited [2]. By modeling a single representative unit cell and applying the Bloch-Floquet theorem [17, 18], a dispersion eigenvalue problem in frequency and wavenumbers or propagation constants is obtained. The solution of this eigenvalue problem for a given set of frequencies or propagation constants yields dispersion curves, which describe the wave propagation in the infinite periodic metamaterial structure.

Stop bands emerge as frequency ranges without freely propagating wave solutions. Although a variety of unit cell modeling methods exist, the finite element (FE) method has gained interest for the analysis of wave propagation in complex structures due to its versatility and high modeling flexibility. Two main FE-based unit cell methods are generally applied [2]. The first approach, named the wave finite element method, discretizes the unit cell using the standard FE method and applies Bloch-Floquet boundary conditions to the unit cell boundaries [19]. The second approach, named the Bloch operator or shift cell operator method, imposes a Bloch-Floquet wave solution to the governing equations of motion before FE discretization of the weak form, and applies regular periodicity boundary conditions to the unit cell boundaries [20]. In this work, the former method is of interest, as it allows straightforward unit cell modeling using standard FE software.

For complex structures, the amount of degrees of freedom (DOFs) in the FE unit cell model can become large and the repeated solution of the dispersion eigenvalue problem for either a given set of frequencies or propagation constants rapidly becomes computationally expensive. If the metamaterial design is well-known, a dispersion curve based stop band prediction might still be feasible for one or a few variants within a reasonable time. However, the time required to explore or optimize a metamaterial design can become prohibitively large due to the need for repeated dispersion curve calculations. To accelerate dispersion curve calculations, a variety of model order reduction (MOR) strategies have recently been proposed. These strategies typically aim to reduce the system of equations for the dispersion eigenvalue problem, by reducing the amount of interior and/or boundary DOFs of the unit cell model. One class of approaches focuses on the condensation of the interior DOFs. Static condensation, or Guyan reduction, is only valid in the static case and increasingly inaccurate with increasing frequency. Dynamic condensation at some prescribed (or all) frequencies of interest is more accurate, but only applies when the dispersion eigenvalue problem is solved for given frequencies instead of given propagation constants [21]. Another class of approaches are the modal reduction methods. A reduced order basis for the unit cell model can for instance be formed by calculating a set of wave modes for some prescribed propagation constants [22, 23]. Alternatively, a component mode synthesis approach can be followed. Bloch Mode Synthesis (BMS) methods reduce the interior DOFs of the unit cell using a truncated set of normal modes, while retaining the boundary DOFs [24–26]. The Generalized Bloch Mode Synthesis (GBMS) additionally also reduces the boundary DOFs of the unit cell using an extended basis with boundary modes [21]. The BMS and GBMS are particularly interesting as they allow creating a reduced order unit cell model before and independently of how the dispersion eigenvalue problem is solved, while significantly reducing the dispersion curve calculation time.

In this paper, the BMS and GBMS methods are applied to a realizable and experimentally validated metamaterial plate design [11]. The acceleration of the stop band predictions is analyzed and next exploited in a design optimization routine to allow fast optimization of the metamaterial design in order to obtain a wide stop band in a targeted frequency range. A variety of stop band optimization approaches have already been proposed in recent years [27–32]. In this work, a black-box global optimization approach is followed, in view of refining and optimizing the geometric parameters of a given metamaterial design. Such approaches rely only on the outputs of the objective function for given inputs, without requiring the actual analytic form of the objective function [33]. Although this generally means a higher number of function evaluations, these methods are easy to use and, combined with a suitable MOR approach to reduce the function evaluation time, are eligible for fast metamaterial design optimization.

This paper is organized as follows. In section 2, the applied methodologies for dispersion curve calculations, model order reduction and optimization are described. Section 3 introduces the metamaterial design and compares the full order and reduced order unit cell models as well as their results and performance. In section 4, the optimization problem is defined and the metamaterial design is optimized for a targeted frequency range, with specific constraints. The main conclusions are summarized in section 5.

2 Methodology

In what follows, the dispersion curve calculation for stop band predictions is first recalled in section 2.1. Next, the MOR using the BMS and GBMS approaches is briefly explained in section 2.2. Section 2.3 eventually provides an overview of the metamaterial design optimization procedure.

2.1 Unit cell modeling for dispersion curve calculations

In this work, metamaterial plates are of interest which are 2D periodic by translation according to a rectangular lattice. To predict the wave propagation in such an infinite periodic metamaterial plate, a general rectangular unit cell model with in-plane dimensions $l_x \times l_y$ is considered, which is discretized using the standard FE method (Fig. 1a). Assuming time-harmonic motion with $e^{j\omega t}$ time dependence, with $j^2 = -1$, angular frequency ω and time t , the equations of motion of the undamped FE unit cell model containing n DOFs are in general given by:

$$(\mathbf{K} - \omega^2 \mathbf{M}) \mathbf{q} = \mathbf{f}, \quad (1)$$

with the $(n \times 1)$ DOF vector $\mathbf{q}$ and generalized force vector $\mathbf{f}$ and the $(n \times n)$ mass matrix $\mathbf{M}$ and stiffness matrix $\mathbf{K}$. For a given unit cell, the FE model matrices can be readily extracted from commercial FE software. In this work, SIEMENS SIMCENTER 3D 2020.1 is used to this end. The DOF vector $\mathbf{q}$ is partitioned according to the nodes in the unit cell interior and on the unit cell boundaries (Fig. 1b):

$$\mathbf{q} = [\mathbf{q}_I^T \mathbf{q}_L^T \mathbf{q}_R^T \mathbf{q}_B^T \mathbf{q}_T^T \mathbf{q}_{BL}^T \mathbf{q}_{BR}^T \mathbf{q}_{TL}^T \mathbf{q}_{TR}^T]^T. \quad (2)$$

The same partitioning is applied to $\mathbf{f}$, $\mathbf{M}$ and $\mathbf{K}$.

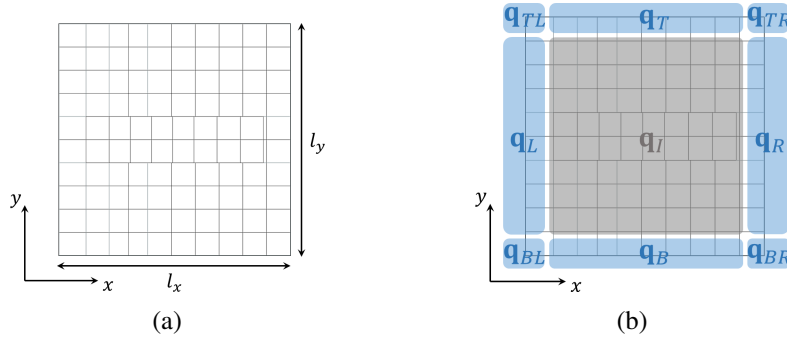


Figure 1: Top view of a rectangular FE unit cell model with in-plane dimensions $l_x \times l_y$, of a structure with 2D in-plane periodicity (a) and partitioned DOF groups on the unit cell boundary and in its interior (b).

The Bloch-Floquet theorem, which governs wave propagation in infinite periodic structures, states that the generalized displacements and forces on the boundaries of the unit cell are scaled with factors $\lambda_x = e^{j\mu_x}$ and $\lambda_y = e^{j\mu_y}$, when moving from one unit cell to the next along the direct lattice vectors of lengths l_x and l_y , respectively [34]. The real and imaginary parts of the propagation constants $\mu_x = k_x l_x$ and $\mu_y = k_y l_y$, with k_x and k_y the structural wave numbers, represent the associated phase and amplitude change, respectively. The Bloch-Floquet theorem allows to formulate the following periodicity boundary conditions, relating $\mathbf{q}$ to the periodic DOF vector $\tilde{\mathbf{q}}$ through the Bloch-Floquet periodicity matrix Λ [19, 21]:

$$\begin{Bmatrix} \mathbf{q}_I \\ \mathbf{q}_L \\ \mathbf{q}_R \\ \mathbf{q}_B \\ \mathbf{q}_T \\ \mathbf{q}_{BL} \\ \mathbf{q}_{BR} \\ \mathbf{q}_{TL} \\ \mathbf{q}_{TR} \end{Bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \lambda_x \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \lambda_y \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \lambda_x \mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \lambda_y \mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \lambda_x \lambda_y \mathbf{I} \end{bmatrix} \begin{Bmatrix} \mathbf{q}_I \\ \mathbf{q}_L \\ \mathbf{q}_B \\ \mathbf{q}_{BL} \end{Bmatrix}. \quad (3)$$

In absence of an external excitation, $\mathbf{f}_I = \mathbf{0}$, the Bloch-Floquet theorem also allows expressing force equilibrium at the unit cell boundaries with its neighboring unit cells in the infinite periodic structure as [34]:

$$\Lambda^H \mathbf{f} = \mathbf{0}, \quad (4)$$

in which $\mathbf{\Lambda}^H$ is the Hermitian transpose of $\mathbf{\Lambda}$. In order for Eqs. (3) and (4) to apply, the nodes on the corresponding unit cell boundaries should be located at the corresponding positions on either side. Combining Eqs. (1)-(4) results in the following dispersion eigenvalue problem in ω and (μ_x, μ_y) [19, 21]:

$$\mathbf{\Lambda}^H(\mathbf{K} - \omega^2\mathbf{M})\mathbf{\Lambda}\tilde{\mathbf{q}} = \mathbf{0}. \quad (5)$$

Two main approaches are generally applied to solve this problem [2, 19, 35]. The first approach, referred to as the *inverse* or $\omega(\mu)$ -approach, imposes a set of real μ , thus free wave propagation, and solves for ω . The second approach, named the *direct* or $\mu(\omega)$ -approach, instead imposes a set of real ω , thus time-harmonic wave propagation, and solves for μ . The direct approach is mainly of interest when analyzing wave attenuation and, in particular, damped unit cells. To predict stop bands, the inverse approach is generally applied to undamped unit cells: dispersion curves for free wave propagation are calculated by imposing real (μ_x, μ_y) along the so-called irreducible Brillouin contour (IBC) [18] and solving Eq. (5) to real ω . Stop bands are then found as frequency ranges in which no freely propagating wave solutions are obtained along the entire IBC. Since mainly the stop band behavior is of interest in this work, the inverse approach is applied to undamped unit cells. The method is implemented in MATLAB R2020A and Eq. (5) is solved using `eigs`.

2.2 Model order reduction

To obtain the dispersion curves, Eq. (5) needs to be repeatedly solved for an imposed set of (μ_x, μ_y) -pairs along the entire IBC. For complex unit cells, however, the FE unit cell model can become large and solving the dispersion eigenvalue problem becomes computationally expensive. To enable fast dispersion curve calculations and thus stop band predictions, in this work, two component mode synthesis techniques are applied to the unit cell FE model: the BMS [24] and the GBMS [21] methods. In what follows, both methods are briefly described, following the theory of [21].

2.2.1 Bloch mode synthesis

The BMS splits the unit cell DOFs in interior and boundary DOFs and replaces the potentially large number of interior DOFs by a small set of modes. This interior modal reduction is performed before applying Bloch-Floquet boundary conditions, which allows obtaining an a-priori reduced order model (ROM) before solving Eq. (5) along the IBC. To this end, the equations of motion in Eq. (1), with $\mathbf{f} = \mathbf{0}$, of the free unit cell model are first partitioned according to the n_I interior DOFs $\mathbf{q}_I$ and n_A boundary DOFs $\mathbf{q}_A$:

$$\left(\begin{bmatrix} \mathbf{K}_{II} & \mathbf{K}_{IA} \\ \mathbf{K}_{AI} & \mathbf{K}_{AA} \end{bmatrix} - \omega^2 \begin{bmatrix} \mathbf{M}_{II} & \mathbf{M}_{IA} \\ \mathbf{M}_{AI} & \mathbf{M}_{AA} \end{bmatrix} \right) \begin{Bmatrix} \mathbf{q}_I \\ \mathbf{q}_A \end{Bmatrix} = \mathbf{0}. \quad (6)$$

Next, a set of fixed interface normal modes Φ_I are calculated to describe the interior unit cell motion:

$$(\mathbf{K}_{II} - \omega_I^2 \mathbf{M}_{II}) \phi_I = \mathbf{0}, \quad (7)$$

$$\Phi_I = \left[\{\phi_{I,1}\}, \{\phi_{I,2}\}, \dots, \{\phi_{I,n_I^\phi}\} \right], \quad (8)$$

with the truncated number of calculated interior modes $n_I^\phi \ll n_I$. In addition, a set of static constraint modes Ψ_{IA} is calculated, to describe the unit cell boundary motion:

$$\Psi_{IA} = -\mathbf{K}_{II}^{-1} \mathbf{K}_{IA}. \quad (9)$$

The unit cell DOFs are now transformed by expressing the interior DOFs $\mathbf{q}_I$ as a linear combination of these fixed interface normal modes Φ_I and static constraint modes Ψ_{IA} :

$$\begin{Bmatrix} \mathbf{q}_I \\ \mathbf{q}_A \end{Bmatrix} = \begin{bmatrix} \overbrace{\Phi_I \quad \Psi_{IA}}^{\mathbf{B}} \\ \mathbf{0} \quad \mathbf{I} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\eta}_I \\ \mathbf{q}_A \end{Bmatrix}, \quad (10)$$

with η_I the reduced set of interior modal DOFs. The transformation matrix $\mathbf{B}$ allows transforming the free unit cell model matrices $\mathbf{K}$ and $\mathbf{M}$ to the reduced BMS unit cell model matrices $\mathcal{K}$ and $\mathcal{M}$:

$$\mathcal{K} = \mathbf{B}^T \mathbf{K} \mathbf{B}, \quad \mathcal{M} = \mathbf{B}^T \mathbf{M} \mathbf{B}, \quad (11)$$

to which the Bloch-Floquet boundary conditions (3)-(4) can subsequently be applied, with Λ updated according to the reduced DOFs. The resulting dispersion eigenvalue problem (5) is straightforwardly solved.

2.2.2 Generalized Bloch mode synthesis

Since the BMS transformation leaves the n_A boundary DOFs $\mathbf{q}_A$ unchanged, for large and complex unit cells, the potentially large n_A can still lead to computationally expensive dispersion curve calculations. Starting from the BMS, the GBMS additionally performs a local interface modal reduction to reduce the boundary DOFs before applying the Bloch-Floquet boundary conditions. To this end, the reduced BMS equations of motion are again first partitioned according to the interior modal DOFs η_I and the boundary DOFs $\mathbf{q}_A$:

$$\left(\begin{bmatrix} \mathcal{K}_{II} & \mathcal{K}_{IA} \\ \mathcal{K}_{AI} & \mathcal{K}_{AA} \end{bmatrix} - \omega^2 \begin{bmatrix} \mathcal{M}_{II} & \mathcal{M}_{IA} \\ \mathcal{M}_{AI} & \mathcal{M}_{AA} \end{bmatrix} \right) \begin{Bmatrix} \eta_I \\ \mathbf{q}_A \end{Bmatrix} = \mathbf{0}. \quad (12)$$

The boundary motion is now described by calculating a set of normal modes Φ_A :

$$(\mathcal{K}_{AA} - \omega_A^2 \mathcal{M}_{AA}) \phi_A = \mathbf{0}, \quad (13)$$

$$\Phi_A = [\{\phi_{A,1}\}, \{\phi_{A,2}\}, \dots, \{\phi_{A,n_A^\phi}\}], \quad (14)$$

with the truncated number of calculated boundary modes $n_A^\phi \ll n_A$. In order for these boundary modes to be suitable for the application of the Bloch-Floquet periodicity boundary conditions, the procedure proposed in [21] is followed. After partitioning the boundary modes Φ_A according to the boundary DOF groups following Eq. (2), the boundary mode partitions of the corresponding DOF groups L and R , B and T as well as BL , BR , TL and TR are respectively combined and orthogonalized using a singular value decomposition before combining them in the boundary transformation matrix $\mathbf{L}_A$, which reduces the entire boundary as:

$$\mathbf{q}_A = \mathbf{L}_A \eta_A, \quad (15)$$

with η_A the reduced set of boundary modal DOFs. Next, the BMS transformation matrix $\mathbf{B}$ of Eq. (10) is extended by incorporating this boundary transformation $\mathbf{L}_A$:

$$\mathbf{B} = \begin{bmatrix} \Phi_I & \Psi_{IA} \mathbf{L}_A \\ \mathbf{0} & \mathbf{L}_A \end{bmatrix}, \quad (16)$$

which now allows transforming the complete set of interior and boundary DOFs $\mathbf{q}$ of the unit cell as follows:

$$\mathbf{q} = \mathbf{B} \eta, \quad (17)$$

with η the entire reduced set of interior and boundary modal DOFs. This transformation matrix $\mathbf{B}$ is eventually used to transform the free unit cell model matrices $\mathbf{K}$ and $\mathbf{M}$ to the reduced GBMS unit cell model matrices following Eq. (11). After updating Λ according to the reduced DOFs, the Bloch-Floquet boundary conditions are straightforwardly applied to the reduced GBMS unit cell model matrices $\mathcal{K}$ and $\mathcal{M}$ and the resulting dispersion eigenvalue problem (5) can eventually be solved. Both the BMS and GBMS approach are implemented in MATLAB R2020A, solving eigenvalue problems (7) and (13) using `eigs`.

2.3 Design optimization

A variety of metamaterial designs have already been practically realized in the past years. However, they are often still tuned in an ad-hoc manner by e.g. manually adapting some geometric parameters of the metama-

terial unit cell model. This can be time-intensive and lead to sub-optimal designs for a targeted frequency range. Therefore, in this work, the aim is to optimize a given metamaterial design in order to obtain an as wide as possible stop band around a targeted frequency, while satisfying specific design constraints.

The problem at hand is treated as a black-box global optimization, which avoids the need to know the precise analytic form of the objective function or the type of design space [33, 36]. To this end, HEEDS 2020.1 (Hierarchical Evolutionary Engineering Design System) [37] is used, which allows performing automated design optimization studies and which allows straightforward interfacing between different modeling software environments. As indicated in the preceding sections, the unit cell is modeled and meshed in SIMCENTER and the MOR and dispersion curve calculations are performed in MATLAB. Within HEEDS, a variety of global optimization algorithms are available, including e.g. genetic algorithm, particle swarm, simulated annealing and direct search methods. Since global optimization algorithms in general require a significant amount of model evaluations, the hybrid, adaptive SHERPA (Systematic Hybrid Exploration that is Robust, Progressive and Adaptive) [36] algorithm is used, which combines elements of multiple search methods and relies on less evaluations as compared to more traditional methods. Contrary to typical static optimization algorithms, SHERPA does not require any tuning parameters to be set prior to execution. Instead, the algorithm learns about the design space and automatically adapts its internal tuning parameters to more efficiently search the design space with fewer evaluations. SHERPA was shown to be capable of competing with and outperforming other global optimization algorithms [36].

The proposed metamaterial design optimization process in HEEDS is shown schematically in Fig. 2. A parametrized unit cell and its FE model in SIMCENTER are fed into HEEDS along with the initial design parameter values and the allowed parameter ranges. The SHERPA algorithm next iterates over a fixed amount of evaluations. SIMCENTER first applies the parameter values to the unit cell, meshes it and exports the model matrices $\mathbf{M}$ and $\mathbf{K}$. Next, $\mathbf{M}$ and $\mathbf{K}$ are fed into MATLAB and reduced to $\mathcal{M}$ and $\mathcal{K}$ using MOR, which are then used to calculate the dispersion curves. Along with the resulting stop band width SB_{width} and center frequency SB_{center} , HEEDS evaluates the objective function and constraints, which are discussed in more detail in section 4.1. The SHERPA algorithm next adapts its internal parameters, determines new design parameter values and the above process repeats until the set amount of evaluations is reached, at which point it terminates. The best feasible design obtained upon termination is the optimized metamaterial unit cell.

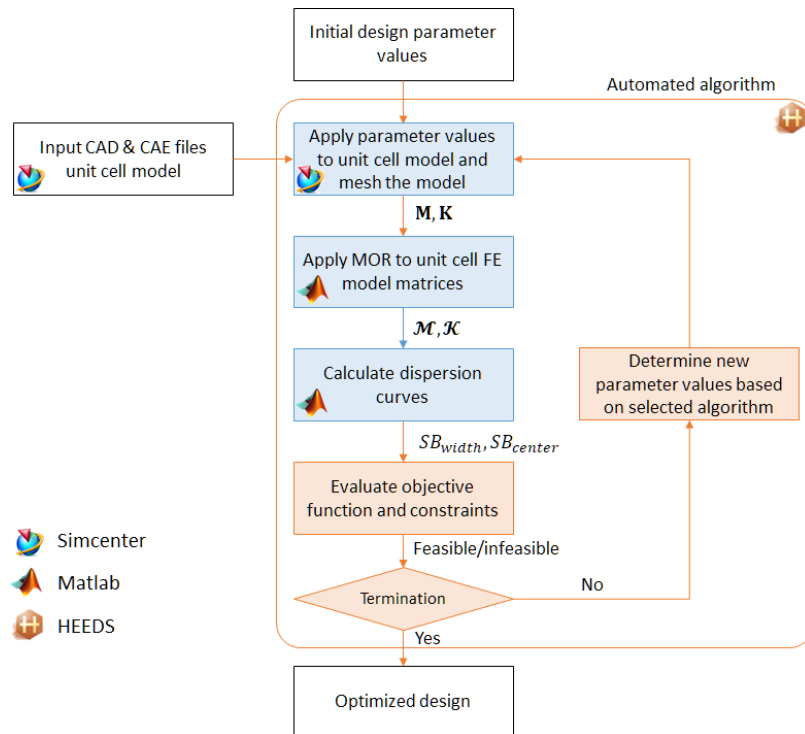


Figure 2: Flowchart of the metamaterial design optimization process in HEEDS [36].

3 Models and performance

After introducing the metamaterial design in section 3.1, the full order model (FOM) and corresponding dispersion curves and stop band behavior are presented in section 3.2. Eventually, the ROMs are discussed and compared with the FOM in section 3.3.

3.1 Metamaterial design

A metamaterial plate is considered, which consists of a 1 mm thick aluminum plate host structure with periodically added beam-shaped PMMA resonators, corresponding to the design in [11]. The representative unit cell of this metamaterial design is shown in Fig. 3. The geometric parameter values for the host structure and resonator are listed in Table 1. The properties of the host structure and resonator materials are shown in Table 2. This metamaterial design aims to achieve a stop band for the acoustically relevant bending waves by means of the first out-of-plane bending mode of the resonator. To this end, the resonators are tuned to a resonance frequency $f_{res} = 616$ Hz and added with a 0.03×0.03 m sub-wavelength spacing to the host structure, leading to a relative added mass ratio m_{ratio} of 39.8%.

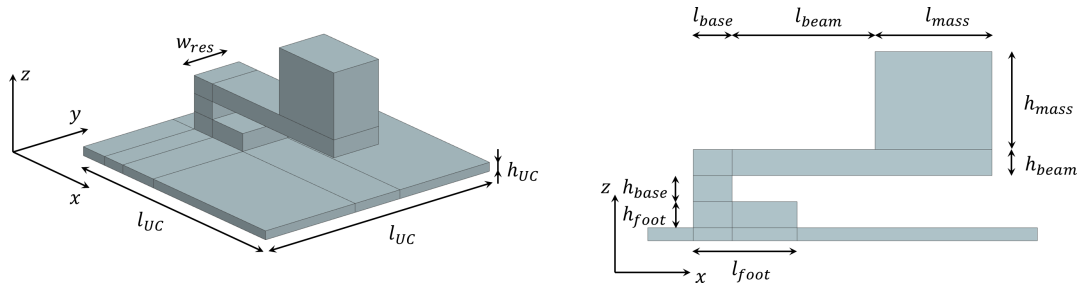


Figure 3: Metamaterial unit cell, with geometric parameter values specified in Table 1.

Table 1: Geometric parameter values of the metamaterial unit cell.

	l_{UC}	h_{UC}	w_{res}	l_{foot}	h_{foot}	l_{base}	h_{base}	l_{beam}	h_{beam}	l_{mass}	h_{mass}
length [mm]	30	1	6	8	2	3	2	11	2	9	7.5

Table 2: Properties of the host structure and resonator materials.

	Young's modulus [GPa]	mass density [kg/m ³]	Poisson's ratio [-]
aluminum	70	2700	0.3
PMMA	4.85	1188	0.31

3.2 Full order unit cell model

The unit cell is discretized using 3834 CHEXA8 linear solid elements, imposing a maximum element size of 1 mm and a minimum of three elements through the thickness of all parts. The FE model contains 16335 DOFs, with $n_I = 14871$ and $n_A = 1464$. The number of DOFs in $\tilde{\mathbf{q}}$ after application of Bloch-Floquet boundary conditions is 15591, which corresponds to the size of the FOM dispersion eigenvalue problem. To analyze the stop band behavior, dispersion curves are calculated along the IBC of the unit cell ($\text{Re}(\mu_x, \mu_y) : O, A, B, C, O \mapsto (0, 0), (\pi, 0), (\pi, \pi), (0, \pi), (0, 0)$ (Fig. 4a)), by solving Eq. (5) in 801 (μ_x, μ_y) -pairs. For every (μ_x, μ_y) -pair, the first five wave solutions are solved, which suffices to find the low-frequency bending wave stop band in the frequency range of interest up to $f_{max} = 2500$ Hz. The latter approximately corresponds to the sub-wavelength limit frequency $f_{\lambda/2} = 2689$ Hz of the bending waves for this unit cell size, above which resonance-based bending wave stop bands are no longer guaranteed [38].

The resulting dispersion curves along the IBC are shown in Fig. 4b. The color indicates the out-of-plane nature of the waves in the host structure, which is calculated using the ratio of out-of-plane to in-plane nodal displacements for the different wave mode solutions: red indicates out-of-plane bending waves and blue indicates in-plane waves. A bending wave stop band is obtained between 597.47 – 656.14 Hz, around f_{res} , which leads to a $SB_{width} = 58.67$ Hz and $SB_{center} = 626.8$ Hz. A computation time of 2024 s is required for the calculation of these dispersion curves along the IBC. All calculations are performed on a laptop with a 2.7 GHz Intel Core i7-4800MQ processor and 16 GB of memory.

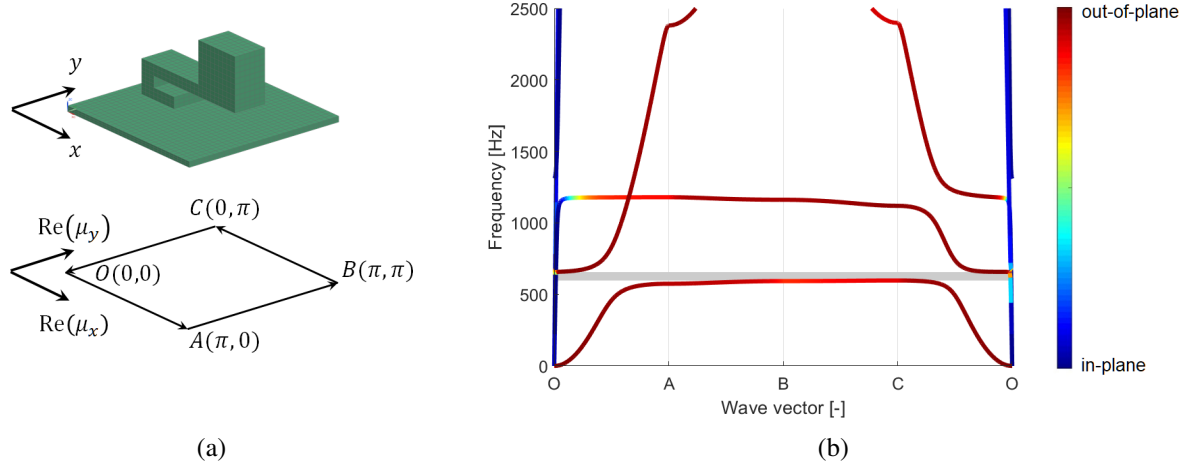


Figure 4: FE model of the metamaterial unit cell with definition of the IBC in the real propagation constant space (a) and dispersion curves along the IBC with the bending wave solutions colored red and bending wave stop band shaded gray (b).

3.3 Reduced order unit cell models

Using the FOM, the dispersion curve calculations take a significant amount of time. The largest number of DOFs is contained within the unit cell interior. To reduce the unit cell interior DOFs, the BMS approach is applied and the number of fixed interface interior modes n_I^ϕ is determined using a frequency-based mode selection by including the modes up to $2f_{max} = 5000$ Hz in the modal reduction basis [24]. This results in $n_I^\phi = 5$ interior modes, which eventually leads to 725 DOFs of the reduced unit cell model after application of Bloch-Floquet boundary conditions. Including the interior modal reduction, this BMS reduced order unit cell model results in a strongly reduced total time of 91 s to obtain the dispersion curves along the IBC. To further reduce this computational time, the remaining boundary DOFs of the unit cell are reduced by applying the GBMS approach. Based on several trials, the first $n_A^\phi = 3n_I^\phi$ boundary modes were found to provide sufficiently accurate results. After making the boundary modes compatible, this $n_A^\phi = 15$ reduces the number of DOFs after application of Bloch-Floquet boundary conditions to only 56. This model reduction further lowers the total time required to obtain the dispersion curves to 15 s. The number of DOFs after application of Bloch-Floquet boundary conditions and the total computational times are shown in Table 3.

Table 3: Comparison of the amount of DOFs after application of Bloch-Floquet boundary conditions and the total computation time (including model reduction for BMS and GBMS) to obtain the dispersion curves along the IBC.

	FOM	BMS	GBMS
DOFs	15591	725	56
total time [s]	2024	91	15

The computational times for the FOM and ROMs are shown in more detail in Fig. 5. For the BMS approach, the interior modal reduction only requires a limited amount of time. The remaining unchanged amount of

boundary DOFs leads to a computation time of the dispersion curves which can still be further improved. For the GBMS approach, the interior modal reduction time is the same as for the BMS approach, while the subsequent boundary reduction for the smaller amount of boundary DOFs is evidently faster. The result is a very fast dispersion curve calculation along the IBC, which only takes a fraction of the total required time for this approach.

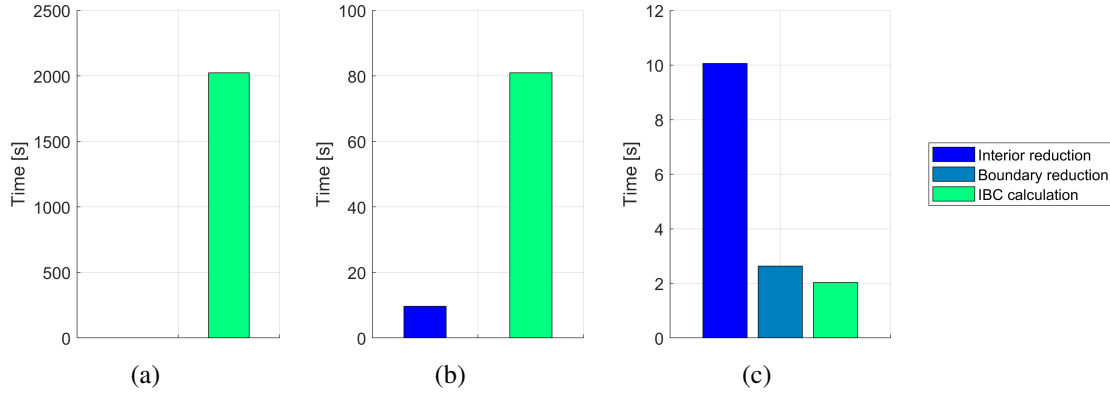


Figure 5: Time required for the IBC calculation and model reduction steps for the full order unit cell model (a) and for the BMS (b) and GBMS (c) reduced order unit cell models.

To also assess the accuracy of the ROM-based dispersion curve calculations, the obtained wave solutions in the frequency range of interest up to f_{max} are compared (Fig. 6). Apart from a slight shift upwards in frequency at higher frequencies near f_{max} , good agreement is obtained between the FOM and ROM solutions. In addition, the relative error between the FOM and ROM solutions on the resulting wave mode frequencies is calculated for every considered (μ_x, μ_y) -pair and wave solution up to f_{max} as:

$$\epsilon = \left| \frac{\omega_{ROM} - \omega_{FOM}}{\omega_{FOM}} \right| \quad (18)$$

and compared (Fig. 7). With the applied $n_I^\phi = 5$ for the BMS, a relative frequency error below 10^{-2} is obtained in the frequency range of interest. For the wave solutions around the stop band zone, this error is even considerably lower. Around the point O of the IBC, the long wavelengths cause a slightly increased error. For the GBMS with $n_I^\phi = 5$ and $n_A^\phi = 15$, the relative frequency error of the wave solutions of interest shows an overall increase, but still remains limited to around 10^{-2} , except for the very low frequency solutions near the point O . The overall reasonably low error is also reflected in the predicted stop band widths and center frequencies for the different models, which show very good agreement (Table 4). The latter is in particular interesting when aiming to obtain fast predictions of the sub-wavelength and typically low-frequency stop bands of metamaterials in the initial stages of their design and analysis.

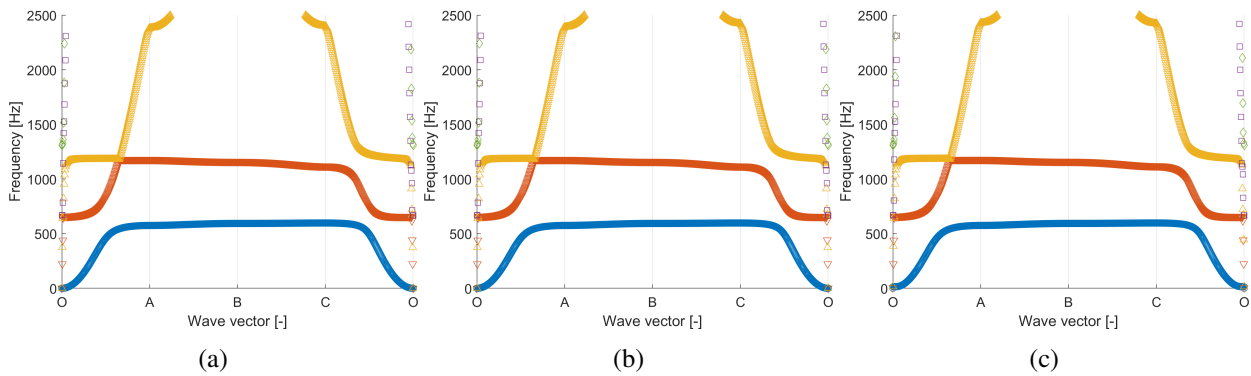


Figure 6: First five wave solutions below 2500 Hz along the IBC, calculated using the full order (a) and BMS (b) and GBMS (c) reduced order unit cell models.

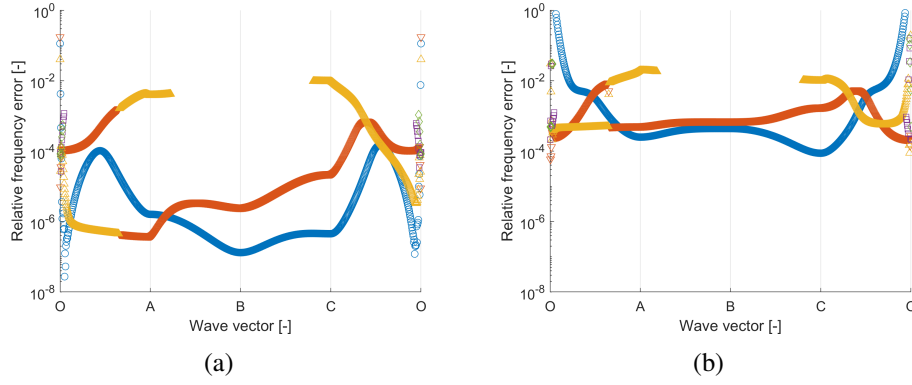


Figure 7: Relative frequency error of the first five wave solutions below 2500 Hz along the IBC for the BMS (a) and GBMS (b) reduced order unit cell models, compared to the full order unit cell model.

Table 4: Comparison of the stop band width and center frequencies for the full and reduced order models.

	FOM	BMS	GBMS
SB_{width} [Hz]	58.67	58.74	58.76
SB_{center} [Hz]	626.80	626.84	626.90

4 Metamaterial design optimization

Since the baseline metamaterial design allows to obtain a stop band, it would be valuable to be able to quickly tune this design to other target frequency ranges. In the preceding section, the GBMS approach was shown to enable fast and accurate stop band predictions, which makes it an appealing approach to leverage upon in metamaterial design optimization. In what follows, the optimization problem at hand is first described in section 4.1. Next, the optimization results are discussed in section 4.2.

4.1 Design parameters, objective function and constraints

In this section, the objective is to create an as wide as possible stop band at lower frequencies than the baseline design of previous section. It is known that mainly the resonator tuning and mass determine the stop band frequency location and width, as long as the resonators are added on a sub-wavelength scale [38]. The resonator geometry will hence be optimized by optimizing its design geometry parameters (Fig. 3), without modifying the host structure and unit cell dimensions, nor their materials. A stop band center frequency between 400 – 450 Hz is aimed for, while the relative mass addition m_{ratio} should not exceed 50% and a spacing of at least 2 mm between consecutive resonators along the x -direction should be maintained to allow proper attachment in practice. The objective function for this design optimization problem is formulated as follows:

$$\text{Maximize } SB_{width}, \quad (19)$$

subjected to:

$$\begin{cases} 400 \text{ Hz} & \leq SB_{center} \leq 450 \text{ Hz}, \\ m_{ratio} & \leq 50\%, \\ 2 \text{ mm} & \leq L_{dif}, \end{cases} \quad (20)$$

with $L_{dif} = l_{UC} - l_{beam} - l_{base} - l_{mass}$ the remaining free space between subsequent resonators along the x -direction. The optimization variables, their baseline values and their ranges are listed in Table 5. The resolution of the design parameter space is 0.5 mm, which approximately corresponds to the accuracy of the laser cutting manufacturing process for the resonators in [11]. The optimization problem is solved using the process described in section 2.3. The unit cell models are reduced by means of the GBMS method with $n_I^\phi = 5$ and $n_A^\phi = 15$, based on the good accuracy obtained for the baseline metamaterial design.

Table 5: Overview of the metamaterial design optimization variables, their ranges and baseline values.

	minimum	baseline	maximum
w_{res} [mm]	2	6	10
l_{base} [mm]	2	3	4
h_{base} [mm]	1	2	4
l_{beam} [mm]	2	11	16
h_{beam} [mm]	1	2	4
l_{mass} [mm]	2	9	16
h_{mass} [mm]	2	7.5	14

4.2 Optimization results

Using the baseline metamaterial design parameters as initial guess for the design variables, the SHERPA algorithm is applied with different predefined numbers of evaluations. The default number of evaluations in HEEDS is 150. In addition, separate optimization runs with 300 and 500 evaluations are also considered to assess the convergence of or difference between the optimized designs as well as the computation time required for a higher number of evaluations. A too low number of evaluations could prevent the optimization algorithm from adequately evolving towards a global optimum and, thus, can reduce the quality of the solution. The latter is, however, sometimes inescapable when only a limited time budget is available, while evaluations are computationally expensive.

The resulting optimized unit cells for the different numbers of evaluations and their corresponding dispersion curves along the IBC obtained after applying the GBMS-approach are shown in Figs. 8 and 9, respectively. The optimal parameter values, stop band widths and center frequencies and constraint parameter values are listed in Table 6. There are some differences between the optimized designs for different numbers of evaluations, yet the resulting stop bands are similar. The optimizer aims to make m_{ratio} as large as allowed and L_{dif} as small as allowed: a longer and heavier resonator allows for a wider and more low-frequency stop band. Although this corresponds to the expected behavior of the first bending mode of a cantilever beam with an end-point mass, such design optimization procedure can help revealing which parameters are more relevant when tuning for a targeted frequency range for this and more complex metamaterial designs. The latter is useful to gain more insight in the design, since local minima with very similar performance could exist, as can be observed upon comparing the optimization result for 300 evaluations to the other cases. Nevertheless, already for a limited number of evaluations, a suitable metamaterial design can be obtained.

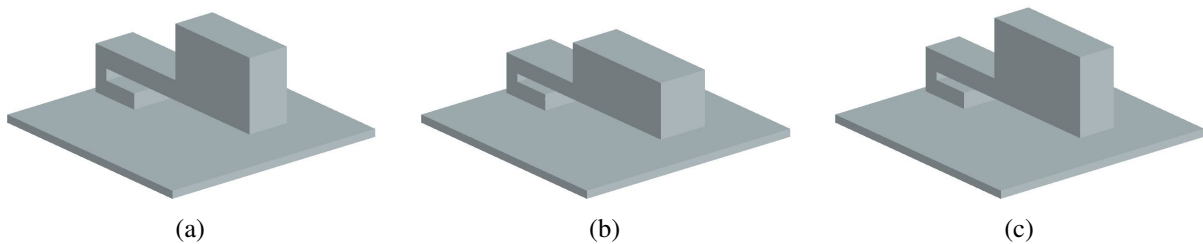


Figure 8: Optimized metamaterial unit cells resulting from the optimization runs using 150 (a), 300 (b) and 500 (c) evaluations.

In order to accurately assess the robustness of the optimization procedure, multiple runs, different initial values and different constraints should also be considered [36]. However, the main objective of this work is to demonstrate that the current optimization approach combined with reduced order unit cell modeling allows for fast metamaterial design optimization. This last point is clearly demonstrated by the computational time required for the different numbers of evaluations (Table 6). The presented procedure allows to obtain a suitable metamaterial design within less than 2 hours, when considering only 150 evaluations. Whereas the SHERPA algorithm aims to reduce the number of required evaluations [36], the application of the GBMS approach in addition successfully reduces the evaluation times. In comparison, 150 evaluations using the full

order unit cell model for the same dispersion curve calculations would require multiple days. It is also noted that part of the computation time is spent on model adaptation and meshing in SIMCENTER and additional overhead and processing in HEEDS, on which the application of the model reduction has no influence.

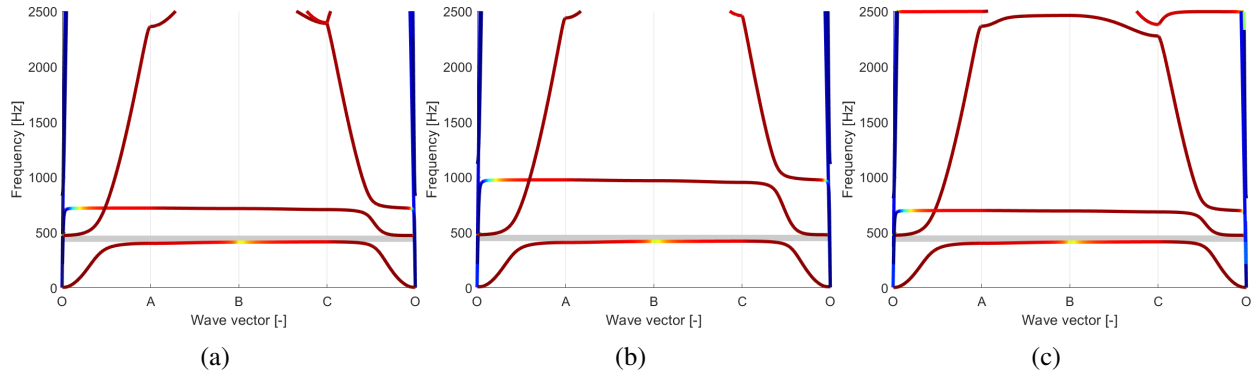


Figure 9: Dispersion curves along the IBC of the optimized metamaterial unit cells resulting from the optimization runs using 150 (a), 300 (b) and 500 (c) evaluations, with stop bands shaded gray.

Table 6: Optimized parameter values, stop band width and center frequencies and constraint parameter values resulting from the optimization runs using 150, 300 and 500 evaluations, as well as the computational time in SIMCENTER and MATLAB and the total computational time.

	150	300	500		150	300	500
w_{res} [mm]	5.5	6.5	5	SB_{width} [Hz]	56.89	57.22	57.91
l_{base} [mm]	2	2	2	SB_{center} [Hz]	443.25	449.67	444.31
h_{base} [mm]	1.5	1	1	m_{ratio} [-]	48.8%	48.3%	49.4%
l_{beam} [mm]	12.5	10	10.5	L_{dif} [mm]	2	2	2
h_{beam} [mm]	2.5	2	2.5	SIMCENTER time	0h56min32s	1h58min05s	3h19min17s
l_{mass} [mm]	13.5	16	15.5	MATLAB time	0h50min58s	1h50min05s	3h37min57s
h_{mass} [mm]	7	5	7.5	total time	1h50min37s	3h55min01s	7h08min51s

Since the n_f^ϕ and n_A^ϕ have been determined using the baseline design only, the optimized stop band widths and center frequencies for the different numbers of evaluations are verified using the corresponding full order unit cell models (Table 7). As expected based on the comparison in section 3.3, good agreement is obtained between the full order and GBMS reduced order unit cell model based stop band predictions. Given the relatively short design optimization throughput time, the result from the first design optimization can serve as a good initial guess for subsequent optimization steps to further fine-tune the result with more precise constraints. The fast optimization achieved by application of the GBMS method, together with the good accuracy of the resulting optimized designs demonstrates the substantial benefits of reduced order unit cell modeling for metamaterial analysis and design in the early metamaterial development stages.

Table 7: Comparison of the stop band width and center frequencies calculated with the full order and GBMS reduced order unit cell models of the optimized metamaterial designs resulting from the optimization runs using 150, 300 and 500 evaluations.

	150		300		500	
	GBMS	FOM	GBMS	FOM	GBMS	FOM
SB_{width} [Hz]	56.89	56.85	57.22	57.20	57.91	57.87
SB_{center} [Hz]	443.25	443.13	449.67	449.53	444.31	444.17

5 Conclusion

In this paper, model order reduction was applied to finite element based metamaterial unit cell models to accelerate dispersion curve calculations. Reduced order unit cell models were obtained by applying two component mode synthesis techniques: the Bloch Mode Synthesis method and the Generalized Bloch Mode Synthesis method. Considerable speed-ups of the dispersion curve calculations along the irreducible Brillouin contour by means of the inverse solution method were demonstrated for a realizable metamaterial design with both methods, with the Generalized Bloch Mode Synthesis method being fastest but slightly less accurate as compared to the Bloch Mode Synthesis method. The resulting frequency solutions and stop band predictions were shown to still be accurate, which is especially interesting for the typical low-frequency range of interest in metamaterial design. The considerable analysis speed-ups with the Generalized Bloch Mode Synthesis were next exploited in a metamaterial design optimization routine, relying on a hybrid adaptive, global optimization algorithm. The resonator geometry was optimized to obtain an as wide as possible stop band in a targeted, low-frequency range, starting from a baseline design. The presented methodology was demonstrated to enable fast metamaterial design optimization as well as to provide insight in the relevant tuning parameters of the metamaterial design. Apart from metamaterials, the applied methodologies could also be used for fast dispersion curve calculations and optimization of a variety of other periodic structures or structures which can be represented by a unit cell model.

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